Probability Generating Function

1 Generating Function

Consider a real sequence $\{a_n\}$ and construct a power series $\sum_{n=0}^{\infty} a_n x^n$. Let R be the radius of convergence of $\sum_{n=0}^{\infty} a_n x^n$, $|x| < R$. Then, the power series can be evaluated finitely, and it's a function of x.

$$\text{Say, } \sum a_n x^n = f(x), \text{ for } |x| < R$$

Then a_n is the coefficient of x^n in the expansion of $f(x), n = 0, 1, 2, \ldots$. The sequence of real no can be generated from the function $f(x) = \sum_{n=0}^{\infty} a_n x^n$. The function $f(x)$ is referred to as the generating function of the sequence.

1.1 Definition

If the power series is convergent on $(-R, R), R > 0$, $\sum_{n=0}^{\infty} a_n x^n = G(x)$ say, $|x| < R$, and $G(x) = \sum_{n=0}^{\infty} a_n x^n$ is called the Generating function of the sequence $\{a_n\}$.

2 Probability Generating Function: (PGF)

2.1 Definition

Let X be a non-negative integer valued random variable with $P(x = k) = p_k$ $k = 0, 1, 2, \ldots$ and $\sum_{k=0}^{\infty} p_k = 1$, then the function $P(t) = \sum_{k=0}^{\infty} p_k t^k = E\left(t^X\right)$, $|t| < 1$, is called the probability generating function of the random variable X.

Remark

1. For, $|t| \leqslant 1, \sum_{k=0}^{\infty} \left|p_k t^k\right| = \sum_{k=0}^{\infty} p_k |t|^k \leqslant \sum_{k=0}^{\infty} p_k = 1$.
   Hence, $\sum_{k=0}^{\infty} p_k t^k = P(t)$ converges absolutely at least for $|t| \leqslant 1$.

2. A pgf uniquely determines a probability distribution.
   Given the PGF $P(t)$ of a random variable with nonnegative integer values, we can determine the probability $p_k = P(X = k)$ for $k = 0, 1, 2, \ldots$. That is why $P(t)$ is called the probability generation function.

3 Examples

1. Suppose a biased coin with probability of head $(= p)$ is tossed repeatedly till the occurrence of 2 successive heads. Suppose x be the no of tosses required.
   Then $E(X) = E(X \mid HH)P(HH) + E(X \mid T)P(T) + E(X \mid HT)P(HT)$.
   $\Rightarrow E(X) = 2p^2 + E(1 + X)q + E(2 + X)pq$.
   Let us replace X by $g(x)$, where $g|x|$ is a real-valued function.

$$E(g(x)) = g(2)p^2 + E(g(1 + x))q + E(g(2 + x))pq.$$

Choose $g(x) = t^x$,

$$\begin{aligned} p(t) &= t^2p^2 + qtp(t) + pqt^2p(t) \\ \Rightarrow p(t) &= t^2p^2\left(1 - qt - pqt^2\right)^{-1} \end{aligned}$$

Though we didn't have the PMF of X, you still have the PGF of X.

Now, expanding this $P(t)$ and collecting the coefficient of t^k, you can get the probability of $P(x = k) = p_k$.

2. Find the PGF of following Distribution.

i) X $\sim$ Bin(n, p)

$$\begin{aligned} P_X(t) = E\left(t^X\right) &= \sum_{x=0}^{n} t^x \binom{n}{x} p^x q^{n-x}, \quad \text{bounded so, convergent.} \\ &= \sum_{x=0}^{n} \binom{n}{x} (pt)^x q^{n-x} \\ &= (q + pt)^n, \quad t \in \mathbb{R} \end{aligned}$$

ii) X $\sim$ Poi(λ)

$$\begin{aligned} P_X(t) &= \sum_{x=0}^{\infty} t^x e^{-\lambda} \frac{\lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(t\lambda)^x}{x!} \\ &= e^{-\lambda} e^{\lambda t} \\ &= e^{-\lambda(1-t)} \\ &= e^{\lambda(t-1)}, \quad t \in \mathbb{R}. \end{aligned}$$

iii) $X \sim$ Geo(p)

$$\begin{aligned} P_X(t) &= E\left(t^X\right) \\ &= \sum_{x=0}^{\infty} t^x pq^x \\ &= p \sum_{x=0}^{\infty} (qt)^x, \quad |qt| < 1 \Rightarrow |t| < \frac{1}{q}; \text{ Radius of convergence } \frac{1}{q} \Rightarrow \left(-\frac{1}{q}, \frac{1}{q}\right) \\ &= p \frac{1}{(1 - tq)} \\ &= p(1 - tq)^{-1} \end{aligned}$$

iv) $X \sim$ NB(r, p)

$$\begin{aligned} P_X(t) &= \sum_{x=0}^{\infty} t^x \binom{x + r - 1}{x} p^r q^x \\ &= p^r \sum_{x=0}^{\infty} \binom{x + r - 1}{x} (tq)^x \\ &= p^r (1 - tq)^{-r} \end{aligned}$$

3. Let X be a RV with PGF P(t), what are PGFs of $Y = a + X, Z = bX$.

   $a) P_Y(t) = E\left(t^Y\right) = E\left(t^{a+X}\right) = t^a E\left(t^X\right) = t^a P_X(t)$.

   If $a < 0$ is an integer, $Y = a + X$ may take negative values and its PGF can be defined in terms of $P_X(t)$.

   $b) P_Z(t) = E\left(t^Z\right) = E\left(t^{bX}\right) = E\left(t^b\right)^X = P_X\left(t^b\right)$.

   If $b = \frac{1}{n}$ for $n \in \mathbb{N}$, then $Z = bX$ can also take fractional values, and its PGF can be defined in terms of $P_X(t)$.

4 PGF of Sum of Independent Random Variables

Let $X_1, X_2 \ldots X_n$ be independent RVs where n is a prefixed positive integer. If $P_{Xi}(t)$ be the PGF of $X_i, i = 1, \ldots, n$.

Then the PGF of $S = \sum_{i=1}^n X_i$ is given by,

$$P_S(t) = \prod_{i=1}^n P_{X_i}(t)$$

Proof. The PGF of S is,

$$\begin{aligned}
&P_S(t) = E\left(t^S\right) \\
&= E\left(t^{X_1+X_2+\cdots+X_n}\right) \\
&= E\left(t^{\sum_{i=1}^n X_i}\right) \\
&= E\left(\prod_{i=1}^n t^{X_i}\right), \quad \text{due to independence} \\
&= \prod_{i=1}^n E\left(t_i\right) \\
&= \prod_{i=1}^n P_{X_i}(t).
\end{aligned}$$

Corollary

If X_i 's are iid with common PGF $P_X(t)$,

$$P_S(t) = [P_X(t)]^n$$

1. If $X_i \sim \text{Bin}\,(n_i, p)$, $X_1 + X_2 \sim$? Hence, $S = \sum_{i=1}^k X_i \sim$?

$$P_S(t) = \prod_{i=1}^k (q + pt)^{n_i} = (q + pt)^{\sum_{i=1}^k n_i}$$

$$S \sim \text{Bin}\left(\sum_{i=1}^n n_i, p\right).$$

2. If $X \sim \text{Bin}(n, p), n \to \infty, p \to 0, np \to \lambda$, show that $X \to \text{Poi}(\lambda)$.

$$\begin{aligned}
P_X(t) &= (q + pt)^n \\
&= (1 - p + pt)^n \\
&= \{1 - p(1 - t)\}^n \\
&= \left\{1 - \frac{\lambda}{n}(1 - t)\right\}^n \\
&\to e^{-\lambda(1-t)}; \quad \text{PGF of Poisson distribution.}
\end{aligned}$$

3. a) $X_i \sim \text{Geo}(p)$, $S = \sum_{i=1}^{r} X_i \sim ?$

   PGF of $X_i = p(1-qt)^{-1}$.

$$P_S(t) = \prod_{i=1}^{r} p_{X_i}(t) = \prod_{i=1}^{r} p(1-qt)^{-1} = p^r(1-qt)^{-r}$$

   $\therefore S \sim \text{NB}(r, q)$

   b) $X_i \sim \text{Poi}(\lambda p)$ then, $S = \sum_{i=1}^{k} x_i \sim ?$

   PGF of $X_i = e^{-\lambda_i(1-t)}$.

$$P_S(t) = \prod_{i=1}^{n} P_{X_i}(t) = \prod_{i=1}^{n} e^{-\lambda_i(1-t)} = e^{-\left(\sum_{i=1}^{n}\lambda_i\right)(1-t)}$$

   $\therefore S \sim \text{Poi}\left(\sum_{i=1}^{n} \lambda_i\right)$

   c) If $X_i \sim \text{NB}\left(r_i, q\right)$, $X_1 + X_2 \sim ?$

$$S = \sum_{i=1}^{k=2} X_i, P_S(t) = \prod_{i=1}^{2} P_{X_i}(t) = \prod_{i=1}^{2} p^{r_1}(1-qt)^{-r_1} = p^{(r_1+r_2)}(1-qt)^{-r_1-r_2}$$

   $\therefore S \sim \text{NB}(r_1 + r_2, q)$

4. If $X \sim \text{NB}(r, q), r \to \infty, q \to 0, rq \to \lambda$, show that $X \to \text{Poi}(\lambda)$.

$$\begin{aligned} P_X(t) &= p^r(1-qt)^{-r} \\ &= (1-q)^r(1-qt)^{-r} \\ &= \left(1-\frac{\lambda}{r}\right)^r \left(1-\frac{\lambda}{r}t\right)^{-r} \\ &= \frac{\left(1-\frac{\lambda}{r}\right)^r}{\left(1-\frac{\lambda t}{r}\right)^r} \to \frac{e^{-\lambda}}{e^{-\lambda t}} = e^{-\lambda+\lambda t} = e^{-\lambda(1-t)} \sim \text{Poi}(\lambda). \end{aligned}$$

5 Calculation of Probability using Generating Function

Method 1

If a RV X has PGF $P_X(t)$, $P[X = n]$ = coefficient of t^n in $P_X(t)$, $\quad \forall n = 0, 1, 2, \ldots$.

Method 2

$$P[X = k] = \frac{1}{k!} \frac{\partial^k P_X(t)}{\partial t^k}\bigg|_{t=0}$$

Proof:

$$P_X(t) = \sum_{n=0}^{\infty} t^n p_n$$

$$\frac{\partial}{\partial t} P_X(t) = \sum_{n=0}^{\infty} n t^{n-1} p_n$$

$$\frac{\partial^2}{\partial t^2} P_X(t) = \sum_{n=0}^{\infty} n(n-1) t^{n-2} p_n = \sum_{n=2}^{\infty} n(n-1) t^{n-2} p_n$$

$$\frac{\partial^k}{\partial t^k} P_X(t) = \sum_{n=0/k}^{\infty} (n)_k t^{n-k} p_n$$

$$\left.\frac{\partial^k P_X(t)}{\partial t^k}\right|_{t=0} = (k)_k t^{k-k} p_k + \left\{ \sum_{n=k+1}^{\infty} (x)_k t^{n-k} p_k \right\}$$
$$= (k)_k p_k.$$
$$\Rightarrow p_k = \frac{1}{k!} \left.\frac{\partial^k P_X(t)}{\partial t^k}\right|_{t=0}$$

6 An important property of PGF

A PGF determines a Probability Distribution uniqualy. Let X and Y be two non-negative integer valued RV such that-

$$p_x = P[X = x] \quad \forall x$$
$$p_y^* = P[Y = y] \quad \forall y$$

Then, the PGF of X is $P_X(t) = \sum_{x=0}^{\infty} t^x p_x$ and the PGF of Y is $P_Y^*(t) = \sum_{y=0}^{\infty} t^y p_y^*$.

Suppose,

$$P_X(t) = P_Y^*(t)$$
$$\Rightarrow \quad \sum_{x=0}^{\infty} t^x p_x = \sum_{y=0}^{\infty} t^y p_y^* \quad \forall |t| \leqslant 1$$

This is an identity in t. So, comparing the coefficient of t^0, t^1,.. ... t^k ..... from both side of identity we get,

$$p_0 = p_0^*, \quad p1 = p_1^*, \ldots, \quad p_k = p_k^*$$

Hence, X and Y have same distribution.

7 Calculation of Moments from PGF

Method 1

If $\mu_{[K]}$ denotes the k th order factorial moment of X, $\mu_{[k]} = \left.\frac{\partial^k P_X(t)}{\partial t^k}\right|_{t=1}$

$$P_X(t) = \sum_{x=0}^{\infty} t^x p_x$$
$$\left.\frac{\partial^k P_X(t)}{\partial t^k}\right|_{t=1} = \sum_{x=k}^{\infty} (n)_k t^{n-k} p_n$$
$$= \sum_{x=k}^{\infty} (n)_k p_n = \mu_{[k]}$$

Method 2

$\mu_{[k]}$ = coefficient of $\frac{(t-1)^k}{k!}$ in $P_x(t)$.

$$P_X(t) = \sum_{n=0}^{\infty} t^n p_n = \sum_{n=0}^{\infty} (1 + t - 1)^n p_n$$
$$= \sum_{n=0}^{\infty} \left\{ \sum_{k=0}^{n} \binom{n}{k} (t-1)^k \right\} p_n. \quad \begin{array}{l} 0 < n < \infty; \\ 0 < k < n \\ \therefore 0 < k < n < \infty, \\ \therefore k < n < \infty \end{array}$$

$$
\begin{aligned}
&= \sum_{n=0}^{\infty}\sum_{k=0}^{n}\binom{n}{k}(t-1)^k p_n \\
&= \sum_{k=0}^{\infty}\sum_{n=k}^{\infty}\frac{1}{k!}(n)_k(t-1)^k p_n \\
&= \sum_{k=0}^{\infty}\frac{1}{k!}(t-1)^k\sum_{n=k}^{\infty}(n)_k p_n \\
&= \sum_{k=0}^{\infty}\frac{(t-1)^k}{k!}\mu_{[k]}
\end{aligned}
$$

8 Problems

1. Matching Problem

$$X = \text{ Number of matches.}$$

$$P[X=m] = \frac{1}{m!}\sum_{r=0}^{n-m}(-1)^r\frac{1}{r!}, m = 0(1)n.$$

The PGF of X is,

$$
\begin{aligned}
P_X(t) &= \sum_{m=0}^{n} t^m p_m \\
&= \sum_{m=0}^{n} t^m\left\{\frac{1}{m!}\sum_{r=0}^{n-m}(-1)^r\frac{1}{r!}\right\} \\
&= \sum_{m=0}^{n}\sum_{r=0}^{n-m} t^m\frac{1}{m!}(-1)^r\frac{1}{r!} \\
s=(r+m), \quad &= \sum_{m=0}^{n}\sum_{s=m}^{n}\frac{t^m}{m!}(-1)^{s-m}\frac{1}{(s-m)!}\frac{s!}{s!} \\
&= \sum_{s=0}^{n}\sum_{m=0}^{s}\frac{1}{s!}\binom{s}{m}t^m(-1)^{s-m} \\
&= \sum_{s=0}^{n}\frac{1}{s!}\sum_{m=0}^{s}(-1)^{s-m}\binom{s}{m}t^m \\
&= \sum_{s=0}^{n}\frac{1}{s!}(t-1)^s
\end{aligned}
$$

$u_{[s]}$ = coefficient of $\frac{(t-1)^s}{s!}$ in $P_X(t)$

$$\mu_{[S]} = 1 \quad \Rightarrow \quad E(X) = \mu_{[1]} = 1$$

$$\mu_{[2]} = 1 \quad \Rightarrow \mathrm{Var}(x) = u_{[2]} + 1 - 1^2 = 1$$

2. A prisoner is trapper in a cell containing 3 doors. The 1st door leads to a tunnel which returns him to his cell after two days of travel. The 2nd after 3 days, and 3rd leads to immediate freedom. Assuming the prisoner will always select the door $1, 2, 3$ with probabilities $\frac{1}{2}, \frac{3}{10}, \frac{11}{15}$. Find the PGF of X, where $X =$ no of days till he returns freedom.

$X =$ no of days taken to get freedom.

$$Y = \text{ The door initially choosen, } Y = \begin{cases} 1 & \text{wp } 1/2 \\ 2 & \text{wp } 3/10 \\ 3 & \text{wp } 1/5 \end{cases}$$

PGF of X is,

$$\begin{aligned} P_X(t) &= E\left(t^X\right) \\ &= E\left(t^X \mid Y=1\right) P(Y=1) + E\left(t^X \mid Y=2\right) P(Y=2) + E\left(t^x \mid Y=3\right) P(Y=3). \end{aligned}$$

$$\begin{aligned} \text{Now,} \quad E\left(t^X \mid Y=1\right) &= E\left(t^{2+X}\right) = t^2 E\left(t^X\right) = t^2 P_X(t) \\ E\left(t^X \mid Y=2\right) &= E\left(t^{3+X}\right) = t^3 E\left(t^X\right) = t^3 P_X(t) \\ E\left(t^X \mid Y=3\right) &= E\left(t^0\right) = 1 \end{aligned}$$

$$P_x(t) = \frac{t^2 P_X(t)}{2} + \frac{3t^3 P_X(t)}{10} + \frac{1}{5}$$

$$P_X(t) = \frac{1/5}{\left(1 - \dfrac{t^2}{2} - \dfrac{3t^3}{10}\right)}$$

$$E(X) = \frac{\partial}{\partial t} P_X(t)\Big|_{t=1} = \left.\frac{\left(1 - \frac{t^2}{2} - \frac{3t^3}{10}\right) \cdot 0 - \frac{1}{5}\left(-\frac{2t}{2} - \frac{9t^2}{10}\right)}{\left(1 - \frac{t^2}{2} - \frac{3t^3}{10}\right)^2}\right|_{t=1} = \frac{19}{2}$$

You can also calculate the probablities.

Alternative,

$$E(X) = E(X \mid Y=1)P(Y=1) + E(X \mid Y=2)P(Y=2) + E(X \mid Y=3)P(Y=3)$$

$$\begin{aligned} E(X) &= E(2+X)\frac{1}{2} + E(3+X)\frac{3}{10} + E(X)\frac{1}{5} \\ \Rightarrow E(X) &= (2+E(X))\frac{1}{2} + (3+E(X))\frac{3}{1} + 0\frac{1}{5} \\ \Rightarrow E(X) &= \frac{19}{2} \end{aligned}$$

9 Generating function of tail function and distribution function

Let X be a RV with PMF $P[X=i] = \begin{cases} p_i, & i = 0, 1, 2, \ldots \\ 0. & otherwise \end{cases}$

Set $p[x_j] = q_j, j = 0, 1, 2 \ldots$, $Q(s) = \sum_{j=0}^{\infty} q_j s^j$, then the series converges for $|s| < 1$ and show that

$$Q(s) = \frac{1 - P(s)}{1 - s}, |s| < 1$$

$P(s)$ is the PGF of X. Find the mean and varience of X (when it exists) in term of $Q(s)$ and its derivative.

Proof:

$$\sum_{j=0}^{\infty} \left|q_j s^j\right| = \sum_{j=0}^{\infty} |s|^j q_j \leqslant \sum_{j=0}^{\infty} |s|^j \quad \text{as } q_j \leqslant 1$$

For $|s| < 1, \sum_{j=0}^{\infty} |s|^j$ is convergent geometric series. By comparison test $\sum_{j=0}^{\infty} q_j s^j$ converges absolutely for $|s| < 1$.

Here $Q(s) = \sum_{j=0}^{\infty} q_j s^j, |s| < 1$ is generating function of the sequerce q_j. Also, $p(s) = \sum_{i=0}^{\infty} p_i s^i$.

Note that,

$$q_{j-1} = p[X > j-1] = p_j + p_{j+1} + p_{j+2} + \dots$$
$$q_j = p[X > j] = p_{j+1} + p_{j+2} + \dots$$

$$\begin{aligned}
& q_{j-1} - q_j = p_j \\
\Rightarrow & \sum_{j=1}^{\infty} q_{j-1} s^j - \sum_{j=1}^{\infty} q_j s^j \quad = \sum_{j=1}^{\infty} p_j s^j \\
\Rightarrow & s \left\{ \sum_{j-1=j'=0}^{\infty} q_{j'} s^{j'} \right\} - \sum_{j=0}^{\infty} q_j s^j + q_0 = \sum_{j=0}^{\infty} p_j s^j p_0 \\
\Rightarrow & s - Q(s) - Q(s) = P(s) - 1 \\
\Rightarrow & Q(s) = \frac{P(s)-1}{s-1} = \frac{1-P(s)}{1-s}, |s| < 1
\end{aligned}$$

$$\begin{aligned}
q_0 &= p[X > 0] \\
&= 1 - p[X = 0] \\
&= 1 - p_0 \\
p_0 + q_0 &= 1
\end{aligned}$$

$$\begin{aligned}
P(s) &= (s-1)Q(s) + 1 \\
P'(s) &= (s-1)Q'(s) + Q(s) \\
P''(s) &= (s-1)Q'(s) + Q'(s) + Q'(s).
\end{aligned}$$

If $E(X)$ exists then $E(X)$ is $P'(1)$,

$$\begin{aligned}
E(X) = Q(1) &= \sum_{j=0}^{\infty} q_j \\
&= \sum_{j=0}^{\infty} p[X > j] \\
&= \sum_{j=0}^{\infty} \{1 - p(X \leqslant j)\} \\
&= \sum_{j=0}^{\infty} \{1 - F_X(j)\}
\end{aligned}$$

If $E\left(X^2\right)$ exists, $E(X(X-1)) = P'(1)$.

$$= 2Q'(1) = 2 \sum_{j=0}^{\infty} q_j j s^{j-1} \Big|_{s=1} = 2 \sum_{j=0}^{\infty} q_J j$$

Exercise

Let $P(S)$ be the PGF of a non-negative integer valued RV, then find the generating function of the following sequences.

i) $q_n = p[X \leqslant n]$
ii) $q_n = p[X < n]$
iii) $q_n = p[X = 2n]$
iv) $q_n = p[x = 3n]$

Solution

iii) $q_n = p[X = 2n] = p_{2n}, \quad n \in \mathbb{N}, n = 0, 1, 2, \dots$

Then the PGF of $\{q_n\}$ is $Q(s) = \sum_{n=0}^{\infty} q_n s^n = \sum_{n=0}^{\infty} p_{2n} s^n$

$$= \sum_{n=0}^{\infty} p_{2n} (\sqrt{s})^{2n}$$

$P(\sqrt{s})^2 = p_0 + p_1\sqrt{s} + p_2(\sqrt{s})^2 + p_3(\sqrt{s})^3 + \ldots$
$P(-\sqrt{s}) = p_0 - p_1\sqrt{s} + p_2(\sqrt{s})^2 - p_3(\sqrt{s})^3 + \ldots$

$$\frac{P(\sqrt{s}) + P(-\sqrt{s})}{2} = p_0 + p_2(\sqrt{s})^2 + p_4(\sqrt{s})^4 + \cdots$$
$$\therefore Q(s) = \frac{P(\sqrt{s}) + P(-\sqrt{s})}{2}$$

iv) $q_n = p[X = 3n]$

$$Q(s) = \sum_{n=1}^{\infty} q_n s^n = \sum_{n=0}^{\infty} q_n (\sqrt[3]{s})^{3n} = \sum_{n=0}^{\infty} p_{3n} (\sqrt[3]{s})^{3n}$$
$$P(\sqrt[3]{s}) = p_0 + p_1 \sqrt[3]{s} + p_2(\sqrt[3]{s})^2 + \cdots$$
$$P(\omega \sqrt[3]{s}) = p_0 + \omega p_1 \sqrt[3]{s} + p_2 \omega^2 (3\sqrt{s})^2 + \cdots$$
$$P\left(\omega^2 \sqrt[3]{s}\right) = p_0 + \omega^2 p_1 \sqrt[3]{s} + p_2 \omega^3 (\sqrt[3]{s})^2 + \cdots$$
$$Q(s) = \frac{1}{3}\left[P(\sqrt[3]{s}) + P[\omega \sqrt[3]{s}) + P\left(\omega^2 \sqrt[3]{s}\right)\right]$$